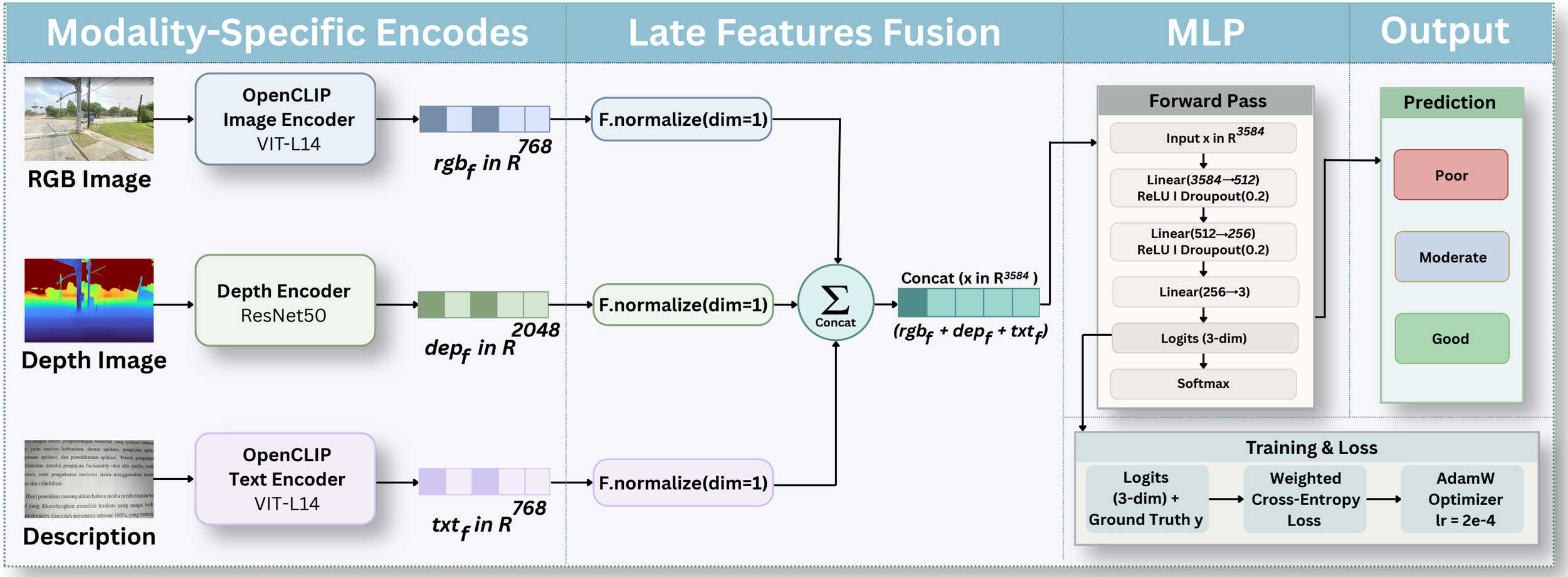
Modality-Specific Encodes
Late Features Fusion
MLP
Output
RGB Image
OpenCLIP Image Encoder VIT-L14
768
rgb_f in R
F.normalize(dim=1)
Depth Image
Depth Encoder ResNet50
2048
dep_f in R
F.normalize(dim=1)
Description
OpenCLIP Text Encoder VIT-L14
768
txt_f in R
F.normalize(dim=1)
Σ Concat
Concat (x in R^3584)
(rgb_f + dep_f + txt_f)
Forward Pass
Input x in R^3584
Linear(3584→512) ReLU I Droupout(0.2)
Linear(512→256) ReLU I Droupout(0.2)
Linear(256→3)
Logits (3-dim)
Softmax
Prediction
Poor
Moderate
Good
Training & Loss
Logits (3-dim) + Ground Truth y
Weighted Cross-Entropy Loss
AdamW Optimizer lr = 2e-4